

CMSE 201
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Fast Fashion and Textile Waste in the United States

This project analyzes historical textile waste and consumption data to understand these trends and predict future landfill waste. Data was collected from the Environmental Protection Agency and a Kaggle dataset .



Background and Motivation

Why This Matters

- Fast fashion encourages frequent clothing purchases
- Cheap, trend-based clothing is often discarded quickly
- This contributes to growing textile waste in landfills
- Understanding this trend can help evaluate long-term environmental impact



Research Question / Data Sources and Variables

Research Question

- How will fast fashion consumption trends affect textile waste in landfills in the United States over time?
- Can historical waste data be used to predict future landfill growth using statistical modeling?

Data

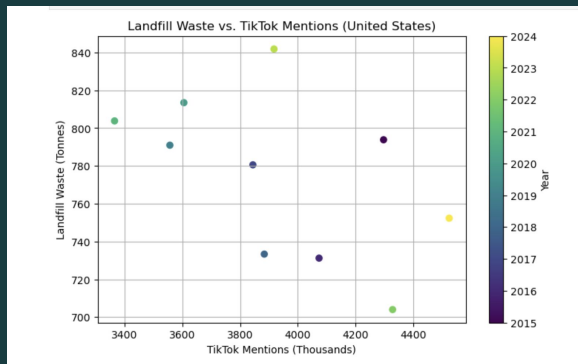
- EPA Dataset
- Kaggle fast fashion dataset

```
1]: fashion.head()
```

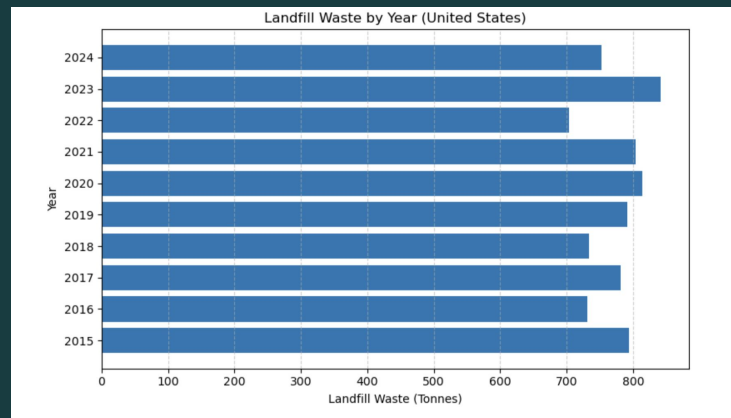
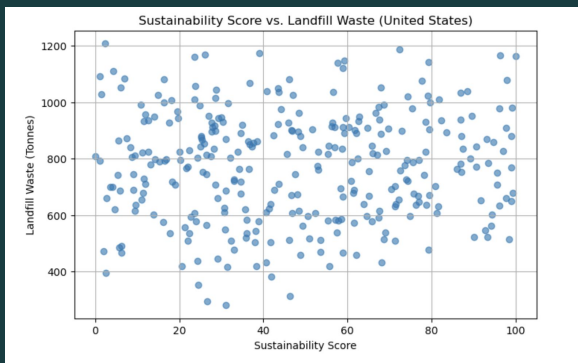
	Brand	Country	Year	Monthly_Production_Tonnes	Avg_Item_Price_USD	Release_Cycles_Per_Year	Carbon_Emissions_tCO2e	Water_Usage_Million_Litres
0	Shein	Indonesia	2017	574.51	19.31	16	11421.58	282.11
1	Forever 21	Vietnam	2024	394.50	9.30	20	5571.01	289.22
2	Uniqlo	India	2024	310.23	25.46	11	10969.00	116.39
3	Forever 21	USA	2017	218.65	13.17	13	13093.00	255.88
4	Zara	Indonesia	2016	1005.84	15.40	20	9548.40	142.07

5 rows x 25 columns

Methodology



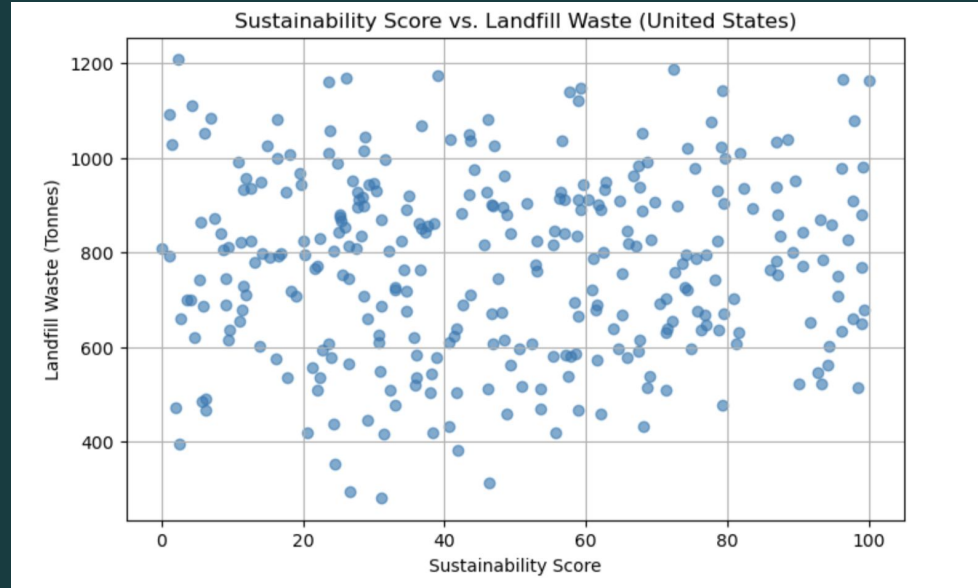
- Created visualizations to examine trends and relationships
- Used correlation analysis to compare variables
- Applied linear regression to model landfill waste
- Built a forecast using historical yearly data



Results: Relationships in the Data

Results: Key Patterns

- Landfill waste generally increased over time
- Production and waste showed a positive relationship
- Waste patterns were influenced by multiple factors
- Social media and sustainability-related variables added complexity



Conclusion and Discussion

Conclusion

- Fast fashion is linked to growing landfill waste
- Landfill waste is shaped by multiple interacting factors
- Historical data can be used to estimate future waste trends
- Long-term environmental impact remains a concern

Limitations

- Limited time range
- Reported data may contain inconsistencies
- Model does not include future policy or behavior changes

